

Class 2

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Today we'll explore the gapminder data. It contains life expectancy, population, and GDP per capita for countries every five years from 1952 to 2007. The gapminder data comes from the Gapminder Foundation: <https://www.gapminder.org/data/> made famous by Hans Rosling's TED talks. We'll continue our exploration using R to understand and visualize data.

Part 1: Getting to Know Gapminder

Make sure you have opened the Rproject first.

Today's data comes packaged inside R itself. Instead of loading a file, we install a package once, then load it with `library()` each session.

Packages are like bonus material for R. We will use them all the time. Installing puts it on your computer. Library opens it (like a book) so everytime you use it you need library in your code first.

```
#install.packages("gapminder") #only run this once ever - then put the # back in front  
library(gapminder)  
data("gapminder") #Puts into our environment
```

```
#install.packages("dplyr") #We'll use this package later, but always a good idea to load packages and d  
library(dplyr)
```

```
##  
## Attaching package: 'dplyr'  
  
## The following objects are masked from 'package:stats':  
##  
##   filter, lag  
  
## The following objects are masked from 'package:base':  
##  
##   intersect, setdiff, setequal, union
```

Looking over the data using commands from yesterday

(hint: use code from yesterday)

```
head(gapminder)
```

```
## # A tibble: 6 x 6  
##   country    continent  year lifeExp      pop gdpPercap  
##   <fct>      <fct>    <int>  <dbl>    <int>    <dbl>  
## 1 Afghanistan Asia      1952   28.8  8425333    779.  
## 2 Afghanistan Asia      1957   30.3  9240934    821.  
## 3 Afghanistan Asia      1962   32.0 10267083    853.  
## 4 Afghanistan Asia      1967   34.0 11537966    836.
```

```
## 5 Afghanistan Asia      1972    36.1 13079460      740.
## 6 Afghanistan Asia      1977    38.4 14880372      786.
```

```
tail(gapminder)
```

```
## # A tibble: 6 x 6
##   country continent year lifeExp      pop gdpPercap
##   <fct>      <fct>    <int>   <dbl>    <int>   <dbl>
## 1 Zimbabwe Africa    1982    60.4  7636524    789.
## 2 Zimbabwe Africa    1987    62.4  9216418    706.
## 3 Zimbabwe Africa    1992    60.4 10704340    693.
## 4 Zimbabwe Africa    1997    46.8 11404948    792.
## 5 Zimbabwe Africa    2002    40.0 11926563    672.
## 6 Zimbabwe Africa    2007    43.5 12311143    470.
```

```
summary(gapminder)
```

```
##           country      continent      year      lifeExp
## Afghanistan: 12 Africa :624 Min. :1952 Min. :23.60
## Albania : 12 Americas:300 1st Qu.:1966 1st Qu.:48.20
## Algeria : 12 Asia :396 Median :1980 Median :60.71
## Angola : 12 Europe :360 Mean :1980 Mean :59.47
## Argentina : 12 Oceania : 24 3rd Qu.:1993 3rd Qu.:70.85
## Australia : 12 Max. :2007 Max. :82.60
## (Other) :1632
##           pop      gdpPercap
## Min. :6.001e+04 Min. : 241.2
## 1st Qu.:2.794e+06 1st Qu.: 1202.1
## Median :7.024e+06 Median : 3531.8
## Mean :2.960e+07 Mean : 7215.3
## 3rd Qu.:1.959e+07 3rd Qu.: 9325.5
## Max. :1.319e+09 Max. :113523.1
##
```

```
dim(gapminder) #Count of how many observations and variables
```

```
## [1] 1704 6
```

```
str(gapminder) #This tells us what kind of data R thinks each variable is - we'll talk about this later
```

```
## tibble [1,704 x 6] (S3: tbl_df/tbl/data.frame)
## $ country : Factor w/ 142 levels "Afghanistan",...: 1 1 1 1 1 1 1 1 1 1 ...
## $ continent: Factor w/ 5 levels "Africa","Americas",...: 3 3 3 3 3 3 3 3 3 3 ...
## $ year : int [1:1704] 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997 ...
## $ lifeExp : num [1:1704] 28.8 30.3 32 34 36.1 ...
## $ pop : int [1:1704] 8425333 9240934 10267083 11537966 13079460 14880372 12881816 13867957 163...
## $ gdpPercap: num [1:1704] 779 821 853 836 740 ...
```

```
length(unique(gapminder$country)) #gives a count of unique countrys
```

```
## [1] 142
```

Checking in (with a partner)

- 1) How many countries are in the dataset?
- 2) What is the average life expectancy?
- 3) What is the median GDP per capita? Average? Bonus:
 - a) Create a scatterplot with the relationship between population and Life Expectancy. Add a trend line

b)What country has the highest life expectancy?

It is best practice to “knit as you go.” Try knitting now to make sure you don’t have errors in your code.

Country and continent are not continuous(numeric in R speak) variables. They are categorical. We can get counts in each category.

frequency tables:

```
table(gapminder$continent) #gives count of different categories
```

```
##  
##   Africa Americas      Asia   Europe Oceania  
##    624      300     396     360      24
```

```
table(gapminder$year)
```

```
##  
## 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997 2002 2007  
##  142  142  142  142  142  142  142  142  142  142  142  142
```

```
prop.table(table(gapminder$continent)) #gives percentages
```

```
##  
##   Africa   Americas      Asia   Europe   Oceania  
## 0.36619718 0.17605634 0.23239437 0.21126761 0.01408451
```

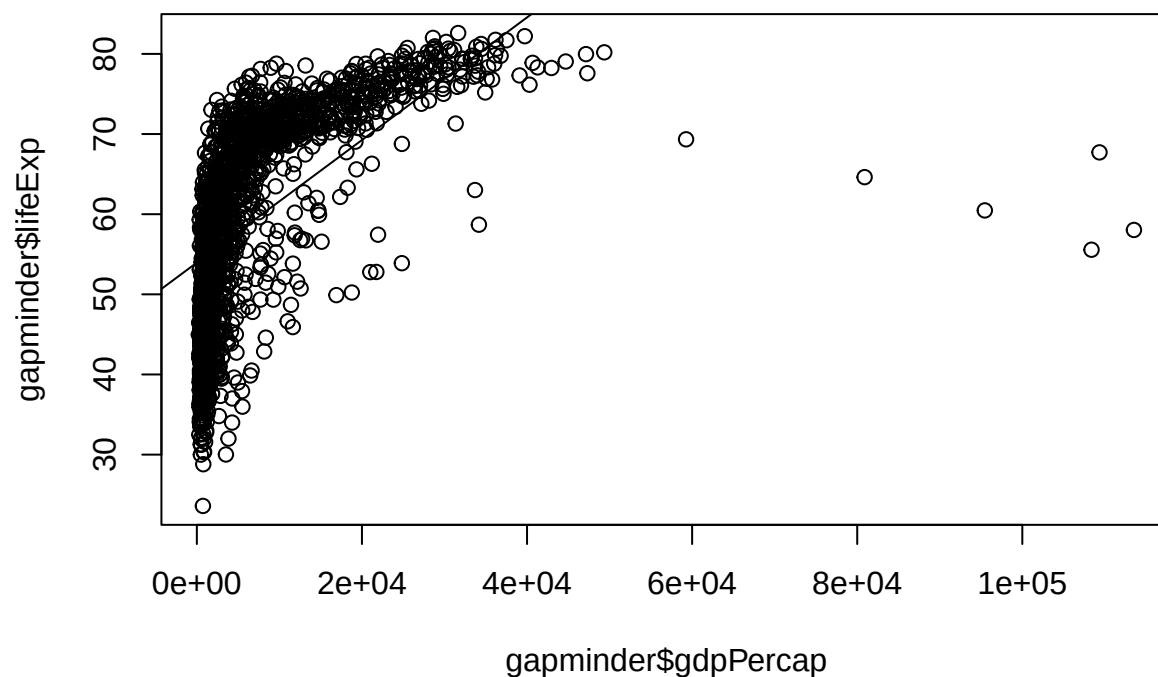
I use this command all the time to get a sense of the data.

Check in: What percentage of the observations in the dataset are from Asia?

“

Plot the relationship between GDP per capita and life expectancy

```
plot(gapminder$gdpPercap, gapminder$lifeExp) #scatterplot (X then Y)  
abline(lm(gapminder$lifeExp~gapminder$gdpPercap)) #add trend line (Y then X)
```



Notice how we always put the explanatory variable on the X axis(horizantal) and the outcome variable on the Y

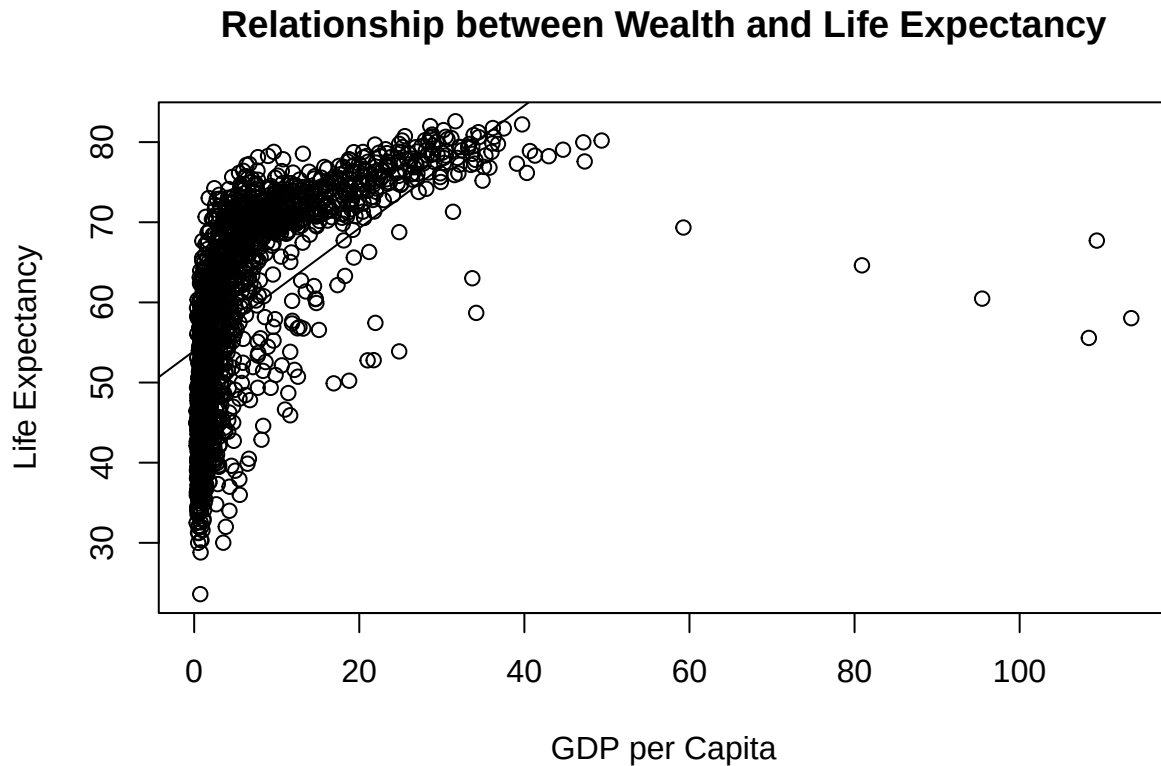
axis (vertical)

GDP per capita hard to read - change unit to thousands

```
gapminder$gdpPercap <- gapminder$gdpPercap/1000
```

Make plot look a bit more professional

```
plot(gapminder$gdpPercap, gapminder$lifeExp, main="Relationship between Wealth and Life Expectancy", xlab="GDP per Capita", ylab="Life Expectancy", abline(lm(gapminder$lifeExp~gapminder$gdpPercap))) #and boom, you've already ran your 2nd regression
```



Does there appear to be a relationship in the sample?

What about with just Asia? There are multiple ways to do this:

We can create a new dataset ie a subset(I almost always choose this option)

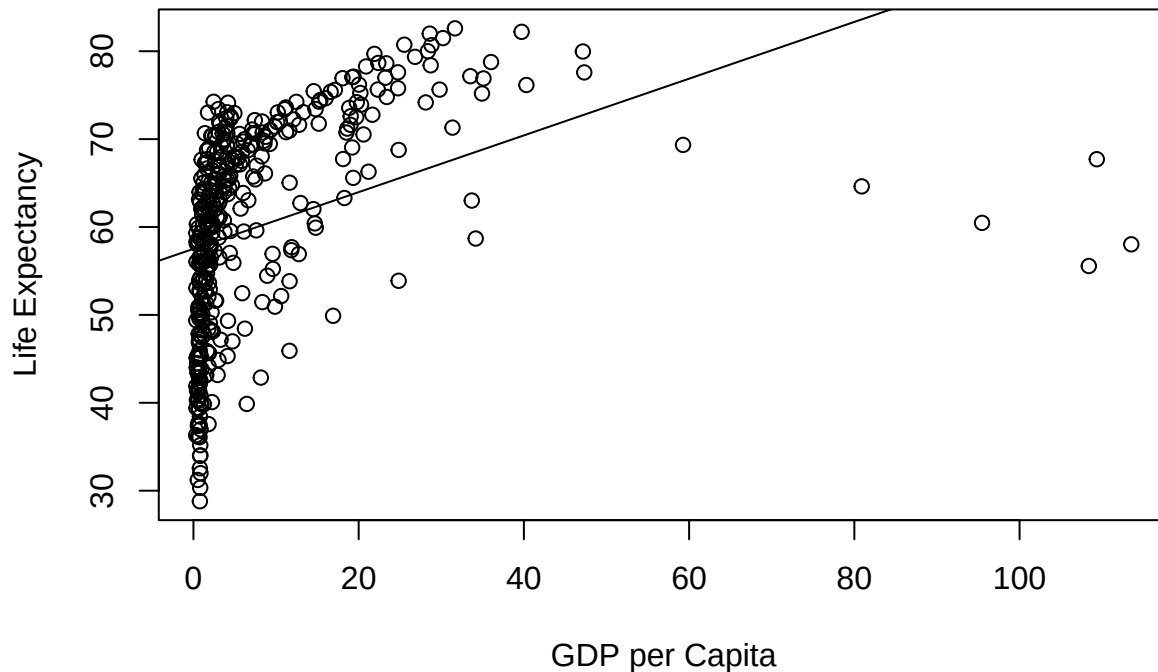
```
table(gapminder$continent)
```

```
##  
## Africa Americas Asia Europe Oceania  
## 624 300 396 360 24
```

```
asia <- gapminder[which(gapminder$continent=="Asia"),]  
asia <- filter(gapminder, continent=="Asia") # This does the same thing using the dplyr package from tidyr
```

```
plot(asia$gdpPercap, asia$lifeExp, main="Relationship between Wealth and Life Expectancy (Asia)", xlab="GDP per Capita", ylab="Life Expectancy", abline(lm(asia$lifeExp~asia$gdpPercap))) #and boom, another regression
```

Relationship between Wealth and Life Expectancy (Asia)



Exclude Europe

```
noteuro <- gapminder[which(gapminder$continent!="Europe"),] # the != does not equal
noteuro <- filter(gapminder, continent!="Europe")
```

Exclude the super rich - use greater than or less than

```
notrich <- gapminder[which(gapminder$gdpPercap < 50),] # less than 50,000 - this drops oil-rich outliers
notrich <- filter(gapminder, gdpPercap < 50)
```

Check in: Try creating a scatterplot with just the 1) Africa OR 2) country-years where life expectancy is less than 70.

#Looking at the data over time: Time to look at something a bit more fun. How has life expectancy changed across the world since the 1950s?

Mean life expectancy?

```
mean(gapminder$lifeExp)
```

```
## [1] 59.47444
```

```
min(gapminder$lifeExp)
```

```
## [1] 23.599
```

```
max(gapminder$lifeExp)
```

```
## [1] 82.603
```

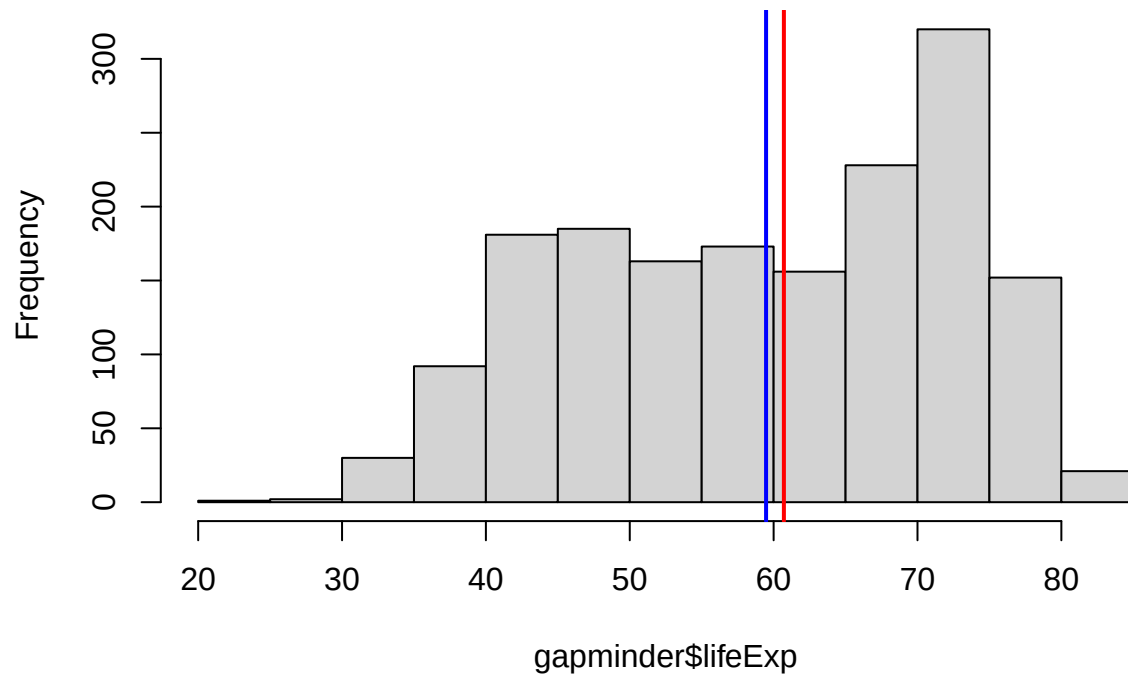
```
range(gapminder$lifeExp)
```

```
## [1] 23.599 82.603
```

Create a histogram for life expectancy

```
hist(gapminder$lifeExp)
abline(v=mean(gapminder$lifeExp), col="blue", lwd=2)#adds a vertical(v) line for the mean
abline(v=median(gapminder$lifeExp), col="red", lwd=2)#a a line for mediant
```

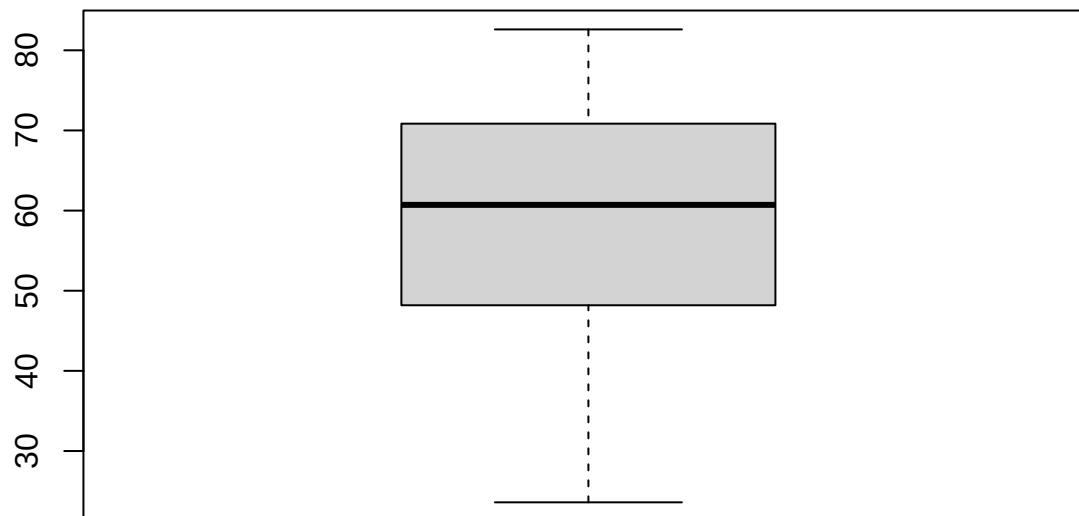
Histogram of gapminder\$lifeExp



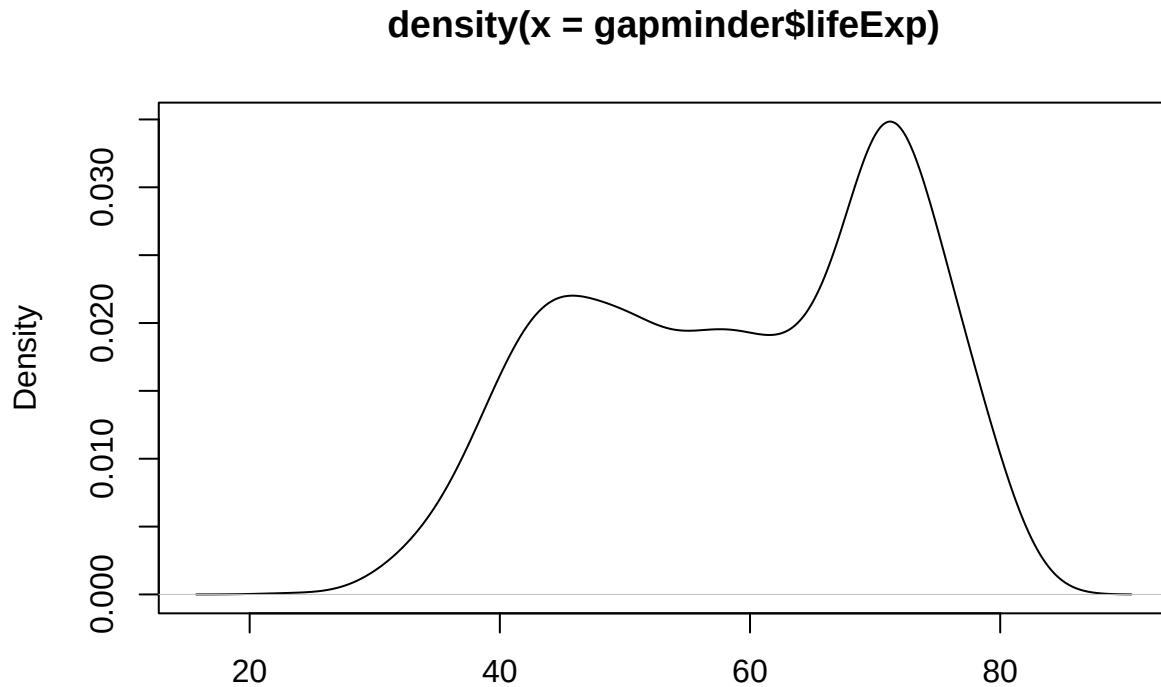
abline gives us a line. v is for vertical

Density plots and boxplots show us similar information

```
boxplot(gapminder$lifeExp)
```



```
plot(density(gapminder$lifeExp))
```



N = 1704 Bandwidth = 2.625

#How can we find out which country-years are at the very bottom?

Check: How many observations are from Africa and how many are from Europe

```
mean(gapminder$lifeExp[gapminder$continent=="Africa"]) #average for Africa
```

```
## [1] 48.86533
```

```
mean(gapminder$lifeExp[gapminder$continent=="Europe"]) #average for Europe
```

```
## [1] 71.90369
```

```
mean(asia$lifeExp)
```

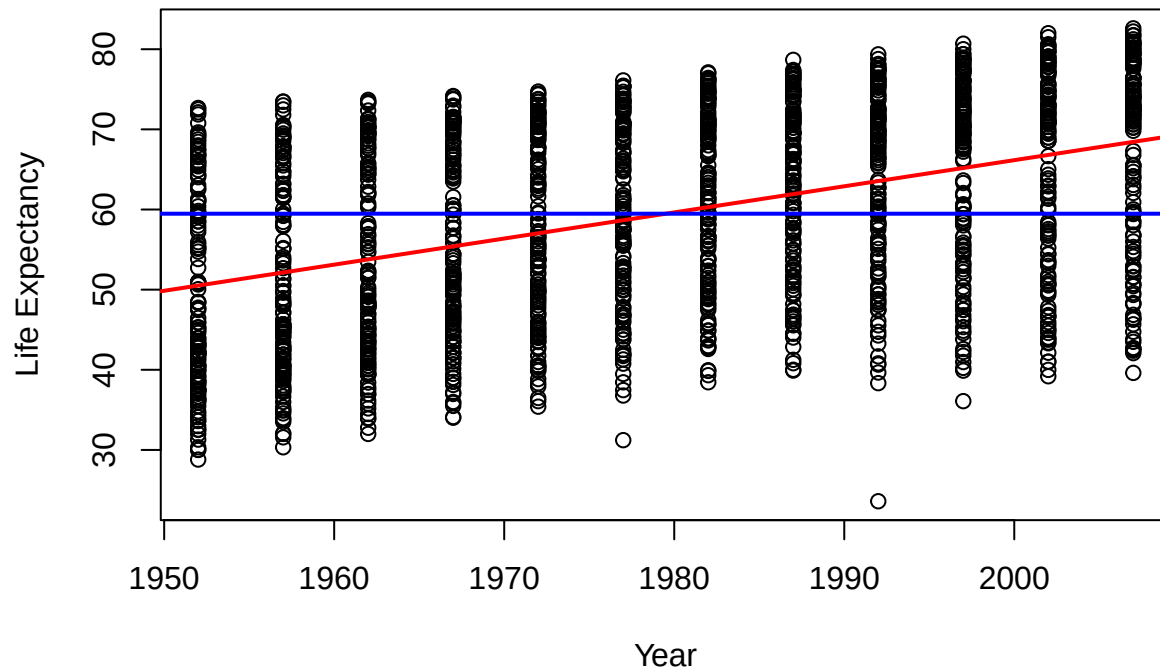
```
## [1] 60.0649
```

#find the mean for the Americas

Let's make a simple scatterplot with year on the horizontal axis, and life expectancy on the vertical axis. A regression allows us to look at the relationship between variables linear regression model, and you're looking at the effect of the IV on the DV

```
plot(gapminder$year, gapminder$lifeExp, ylab = "Life Expectancy", xlab = "Year", main = "Life Expectancy vs Year")
abline(lm(gapminder$lifeExp ~ gapminder$year), col="red", lwd=2) # What type of relationship is this?
abline(h=mean(gapminder$lifeExp), col="blue", lwd=2)
```

Life Expectancy Over Time



That is a bit overwhelming lets just look at 2007 - ie lets create a subset

```
g2007 <- gapminder[which(gapminder$year==2007),] #just using base R
```

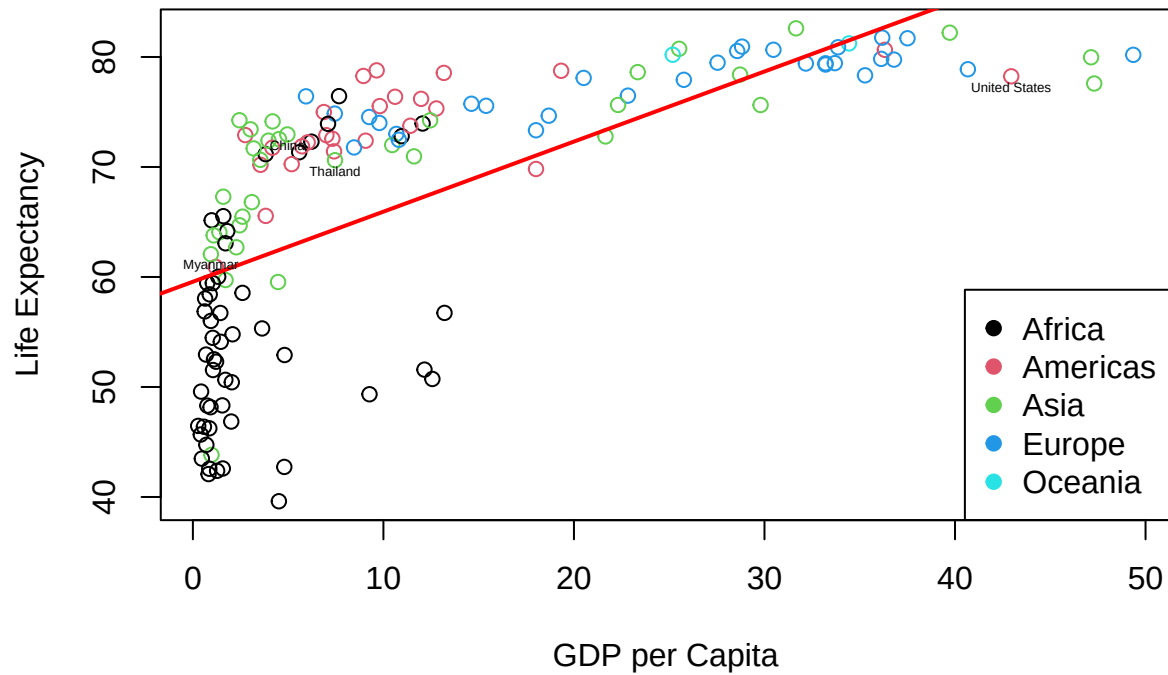
```
#New Graph
```

```
plot(g2007$gdpPercap, g2007$lifeExp, ylab = "Life Expectancy", xlab = "GDP per Capita", main = "Wealth & Life Expectancy")
abline(lm(g2007$lifeExp ~ g2007$gdpPercap), col="red", lwd=2)
legend("bottomright", legend = levels(factor(g2007$continent)), pch=19, col = factor(levels(factor(g2007$continent))))
```

```
#I want to add labels for specific countries
```

```
g2007$stars <- ifelse(g2007$country=="United States" | g2007$country=="China" | g2007$country=="Myanmar" | g2007$country=="Russia", 5, 0)
text(g2007$gdpPercap, g2007$lifeExp-1, g2007$stars, cex=.4) #add text to location
```

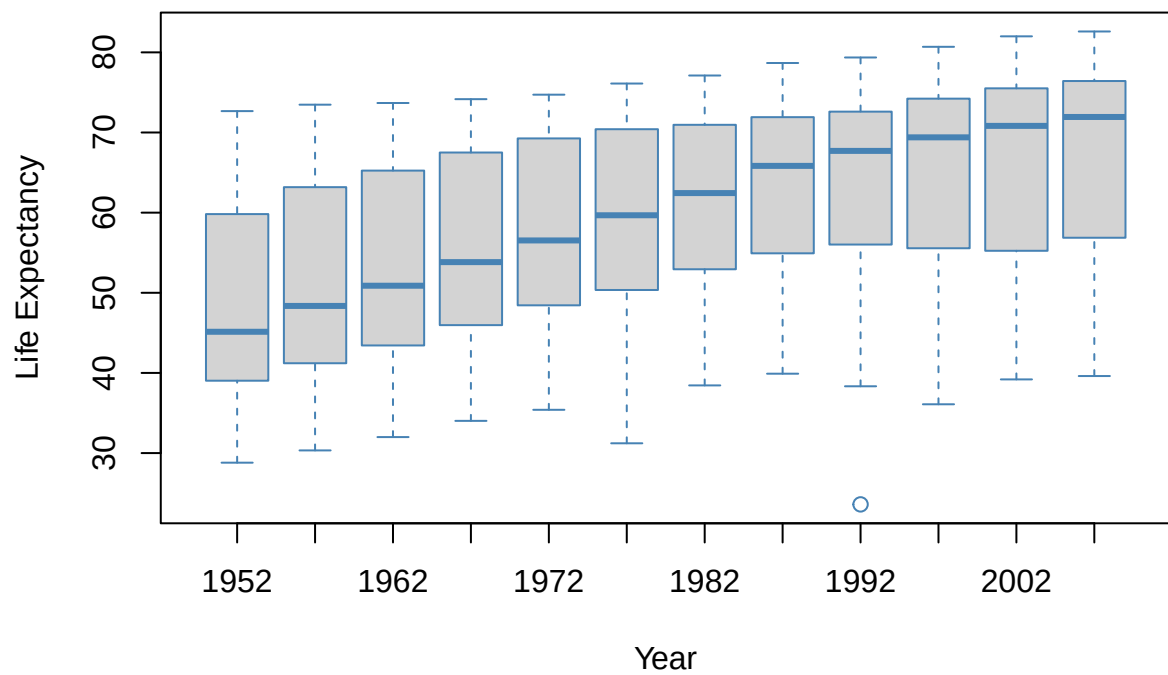
Wealth and Life Expectancy in 2007



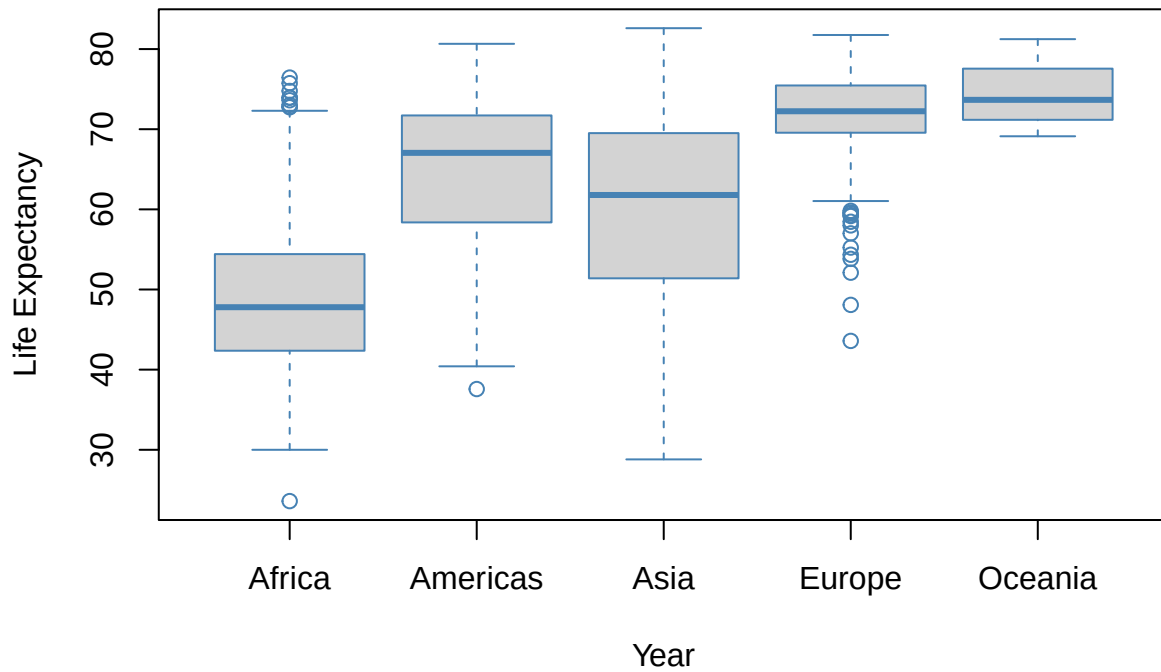
Check: Can you create the same for 1952

A boxplot for each year can show us how the distribution has changed over time

```
boxplot(gapminder$lifeExp~gapminder$year, xlab = "Year", ylab="Life Expectancy", border = "steelblue")
```



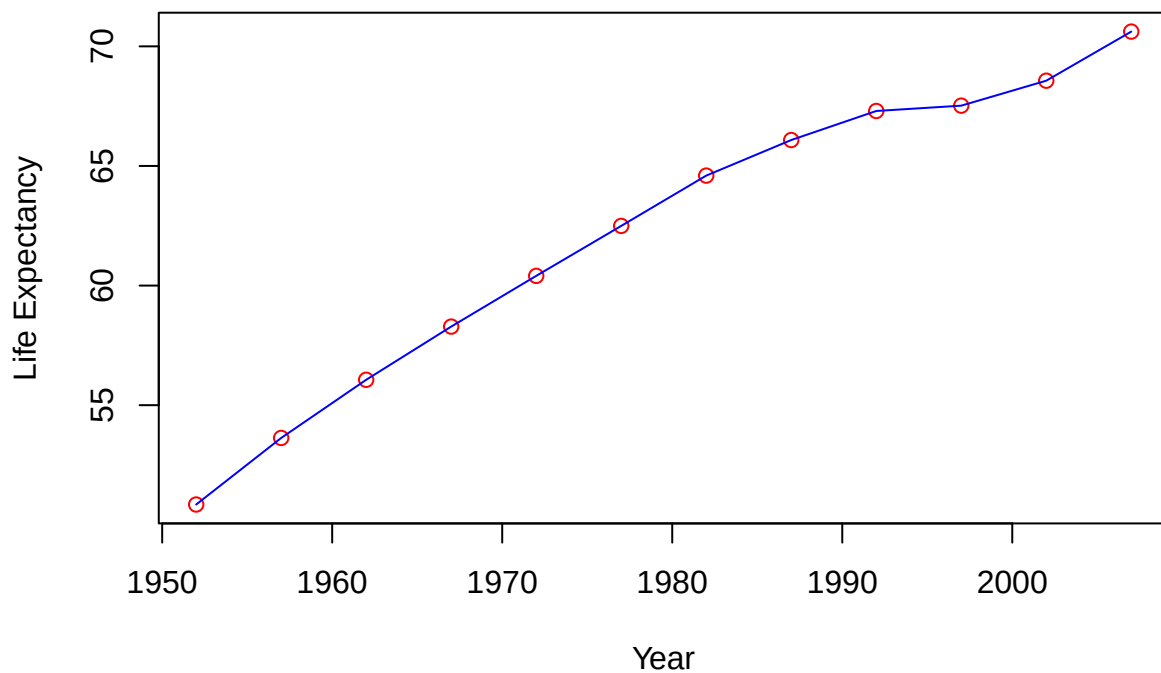
```
boxplot(gapminder$lifeExp~gapminder$continent, xlab = "Year", ylab="Life Expectancy", border = "steelblue")
```



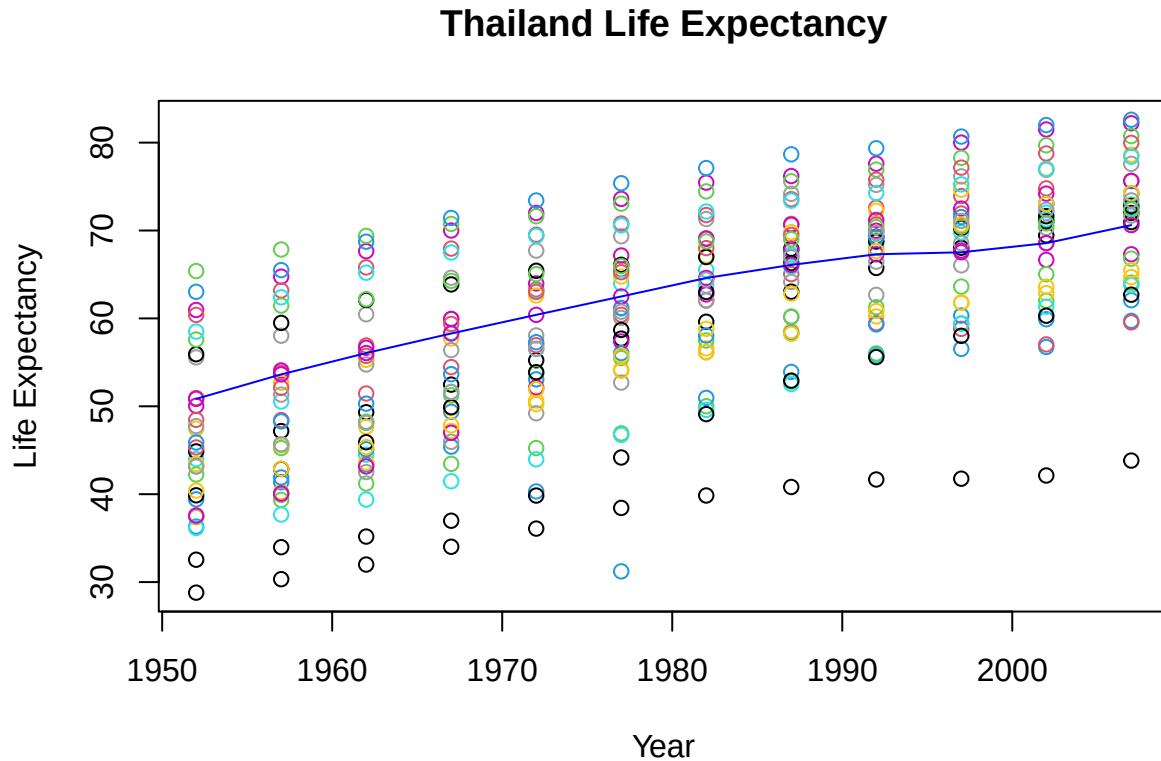
Lets look at changes in thailand over time

```
thai <- filter(gapminder, country=="Thailand")
plot(thai$year, thai$lifeExp, xlab="Year", ylab="Life Expectancy", col="red", main="Thailand Life Expectancy")
lines(thai$year, thai$lifeExp, col="blue")
```

Thailand Life Expectancy



```
plot(asia$year, asia$lifeExp, xlab="Year", ylab="Life Expectancy", col=gapminder$country, main="Thailand Life Expectancy")
lines(thai$year, thai$lifeExp, col="blue")
```



I've clearly had too much fun. Your turn to make something interesting.

Since we've created some new variables we may want to save our data:

```
write.csv(g2007, "g2007.csv") #writes a csv to your working directory that you can open in excel
save(g2007, file="g2007.Rdata") #saves Rdata format to your working directory
```